

DEEP LEARNING: ADVANCED MODELS AND METHODS · DL26

Query-Conditioned Visual-Token Reconstruction

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github.com/SimoneAvellino/VisualTokenRetrieval

PROBLEM

We want to process video online

Process frame-by-frame and discard past frames

What if we need informations about discarded frames?

We use the query to retrieve the past informations.

Memory problems

Keep a long video in memory is not possible. The cost of processing is quadratic.

Ego4D: Retrieving Exact Moments in Hours of Egocentric Video



Natural Language Queries (NLQ) Task

First-person daily-life videos. Each sample links a video, a natural-language question, and a ground-truth time window.

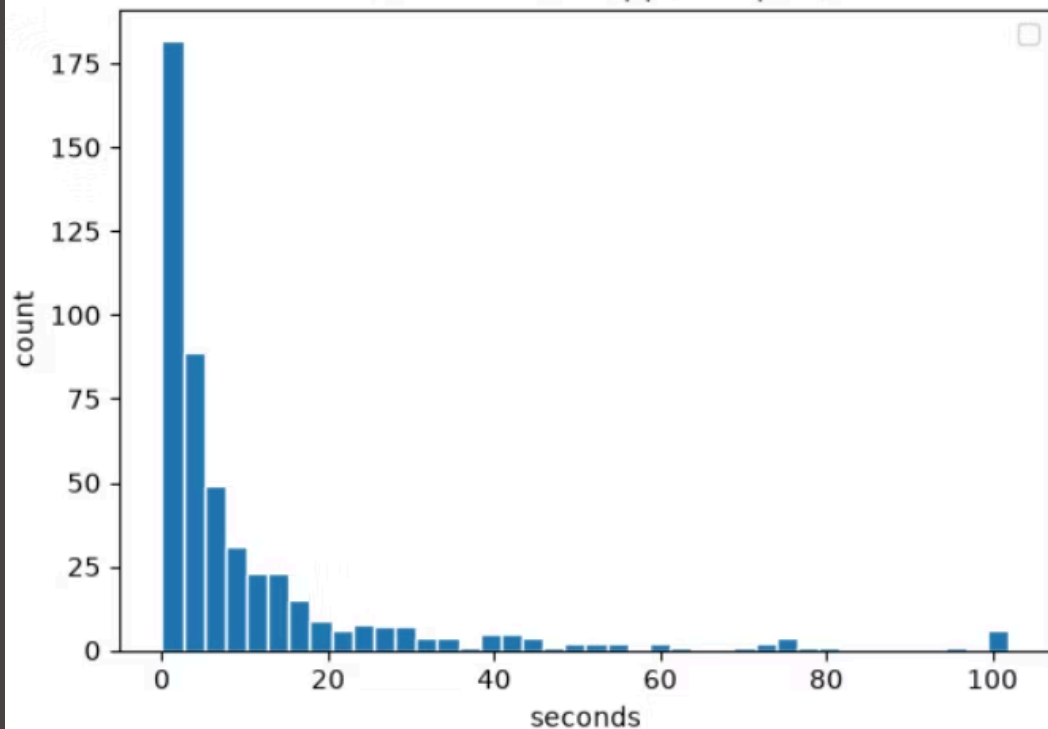
Example: "When did I pick up the cup?" → 45.2–62.5 s

Strict Video-Level Split

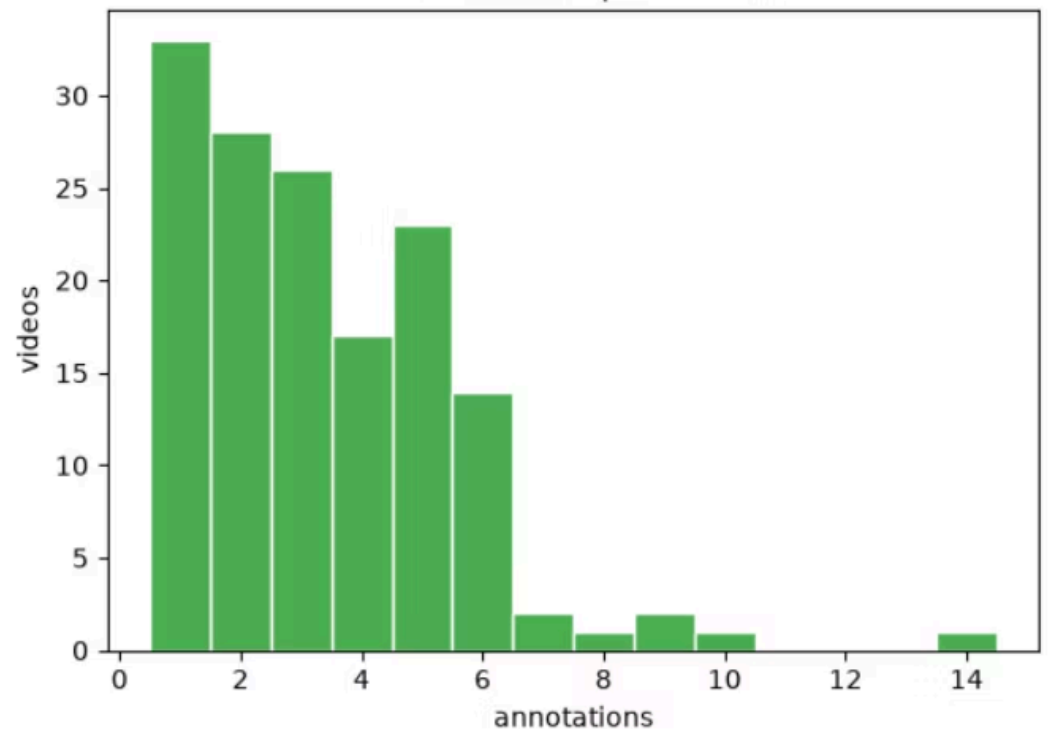
- 348 samples for training
- 68 samples for validation
- 44 samples for testing

⚠ Only 148 videos carried extracted tokens ⚠

GT duration (s, clipped at p99)



Annotations per video



ARCHITECTURE

Baselines

Mean tokens

Returns the mean token of the entire video

Center window

Returns the centered window of the context

Measuring Reconstruction & Retrieval

Reconstruction metrics:

cos_token

Token-by-token average cosine similarity.

norm_ratio

Predicted/true norm ratio. Ideal = 1.0. Detects scale collapse.

Retrieval metrics:

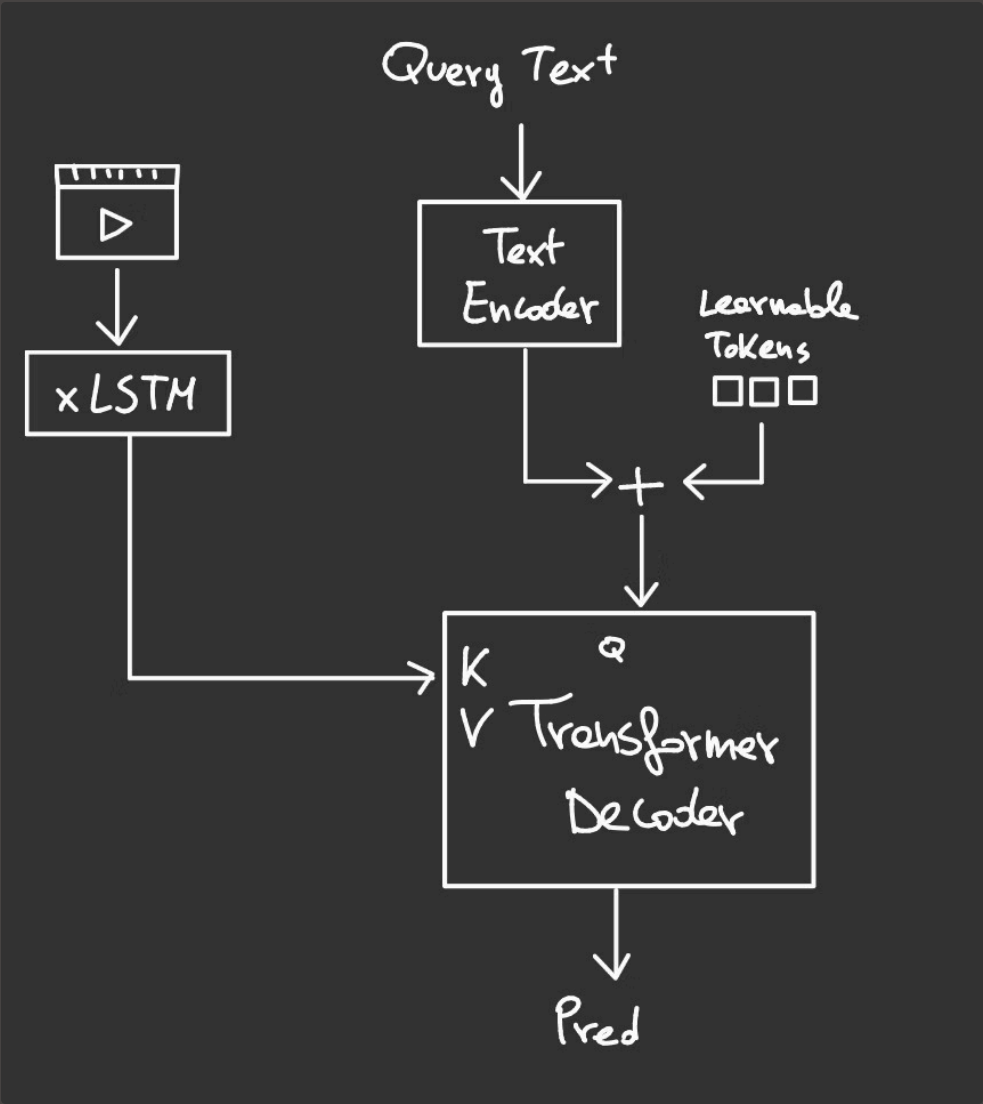
Recall@K

Fraction of predictions where the own GT moment ranks within the top-K most similar candidates.

MRR

Average of $1/\text{rank}$ of the true moment. Higher = the right moment sits near the top.

Mean Token Reconstructor

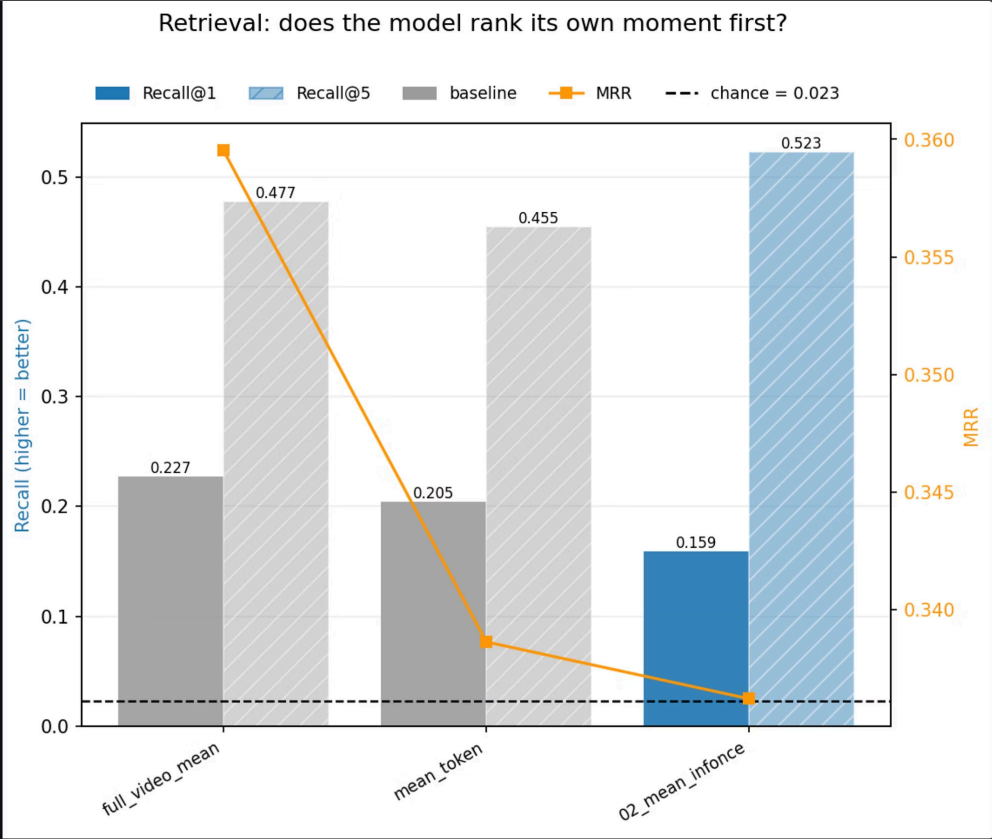


- Video processed by xLSTM → full temporal sequence preserved
- Question (frozen CLIP) guides *where* to look via cross-attention
- Predicts a **single vector** $\approx$ mean of GT window tokens
- Loss: InfoNCE + cosine anchor

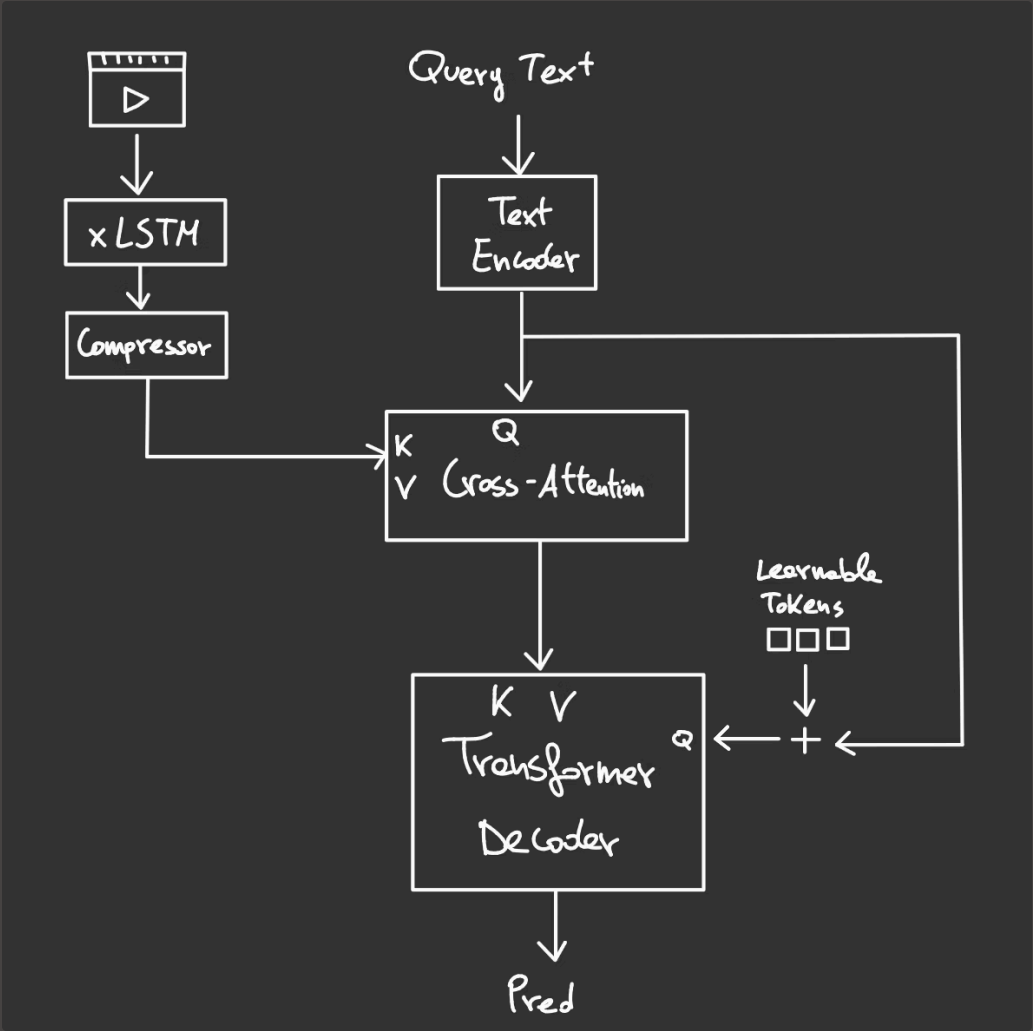
Ablation Study:

- re-run retrieval replacing each sample's question with a *wrong* question

Right query	Wrong query
• Recall@1: 0.1591 • Recall@5: 0.5227 • MRR: 0.3362	• Recall@1: 0.1591 • Recall@5: 0.4773 • MRR: 0.3124

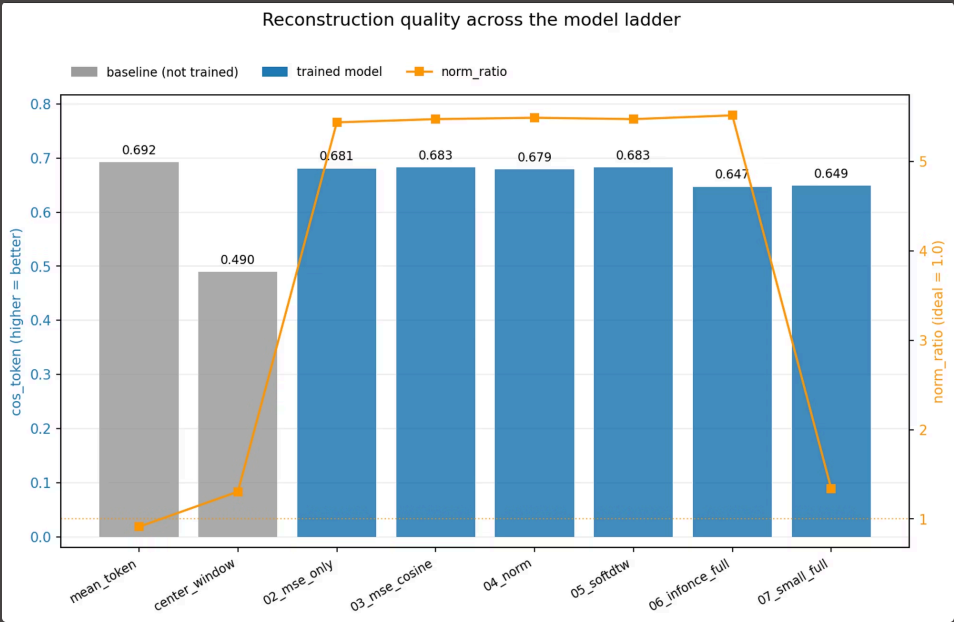


Query-Conditioned Token Reconstructor

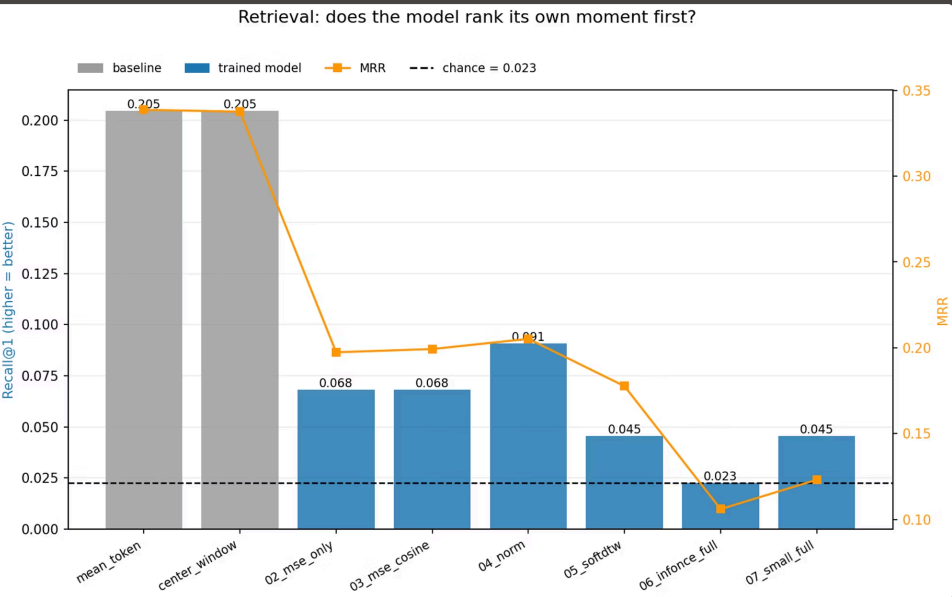


- Video compressed by xLSTM + bottleneck → fixed query set Z
- Text tokens cross-attend $Z \rightarrow$ query-conditioned memory C
- Learnable slots + question summary decode $C \rightarrow$ **token sequence**

Results on reconstruction



Results on the retrieval



RESULTS

Conclusions

- Mean Reconstructor **ignores the query**
- Trained models do not **surpass the baseline**.
- Query-Conditioned Reconstructor learns **good reconstruction** but **retrieval collapses**
- Train on larger dataset
- Improve how in the InfoNCE negatives are retrieved